B. Petr#

time limit per test: 2 seconds

memory limit per test: 256 megabytes

Long ago, when Petya was a schoolboy, he was very much interested in the Petr# language grammar. During one lesson Petya got interested in the following question: how many different continuous substrings starting with the s_{begin} and ending with the s_{end} (it is possible $s_{begin} = s_{end}$), the given string t has. Substrings are different if and only if their contents aren't equal, their positions of occurence don't matter. Petya wasn't quite good at math, that's why he couldn't count this number. Help him!

Input

The input file consists of three lines. The first line contains string t. The second and the third lines contain the s_{begin} and s_{end} identificators, correspondingly. All three lines are non-empty strings consisting of lowercase Latin letters. The length of each string doesn't exceed 2000 characters.

Output

Output the only number — the amount of different substrings of t that start with s_{begin} and end with s_{end}.

Examples

input

```
round
ro
ou
```

output

```
1
```

input

```
codeforces
code
forca
```

output

```
0
```

input

```
abababab
a
b
```

output

```
4
```

input

```
aba
ab
ba
```

output

```
1
```

Note

In the third sample there are four appropriate different substrings. They are: `ab`, `abab`, `ababab`, `abababab`.

In the fourth sample identificators intersect.

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 86 (Div. 1 Only)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
361065086	Feb/02/2026 02:23	Accepted
361064814	Feb/02/2026 02:14	Accepted
361064768	Feb/02/2026 02:12	Time limit exceeded on test 76

→ Problem tags

brute force | data structures | hashing | strings | *2000

No tag edit access